

ROM mapping. After POST, BIOS executes INT 19h, the bootstrap loader service. This routine begins the OS boot process.

Warm vs cold boot

BIOS checks the memory location 0x0472 in the BIOS data area.

- If value = 1234h → Warm boot (restart).
- Otherwise → Cold boot (fresh power-on).

The CMOS RAM contains a shutdown status byte that also helps BIOS determine the reset type.

Boot device selection

BIOS checks devices (HDD, USB, CD, etc) based on the boot order stored in CMOS.

For each device:

- It reads the first sector (512 bytes) = MBR (Master Boot Record).
- If the last two bytes = 0xAA55 → MBR is valid.
- If invalid → BIOS continues to the next boot device.

If no valid boot device is found, BIOS executes INT 18h (no boot device found error).

Master Boot Record (MBR)

Once a valid MBR is found, BIOS loads it into memory at address 0x7C00.

The MBR structure is:

- 446 bytes – Bootloader code
- 64 bytes – Partition table (4 entries × 16 bytes)
- 2 bytes – Boot signature (0xAA55)

Note that MBR has no backup and it can only describe four primary partitions (limitation).

Partition Boot Record (PBR/VBR)

MBR boot code checks the partition table for an active partition (bootable flag = 0x80).

It loads the first sector of the active partition (the Volume Boot Record) into memory at 0x7C00. VBR contains filesystem-specific boot code and metadata:

- Sector size, sectors per cluster
- Partition type (FAT, NTFS, ext, etc)

The VBR boot code loads the second-stage bootloader (GRUB, LILO, etc).

Bootloader stages

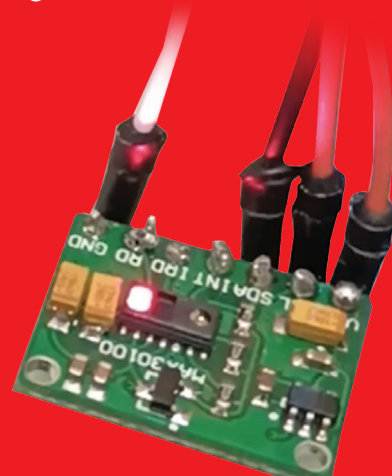
The bootloader is executed in multiple phases.

- *Zero stage bootloader*: Startup code inside BIOS.
- *First stage bootloader (MBR + VBR code)*: Minimal initialisation, loads second stage.
- *Second stage bootloader (GRUB/LILO)*: Provides full boot menu, loads the OS kernel.

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